Proof

We give a combinatorial proof of the theorem.

- When we expand the product $(x+y)^n$, the terms are of the form $x^{n-k}y^k$ for $k = 0, 1, \ldots, n$.
- To obtain a term of the form $x^{n-k}y^k$, it is necessary to choose k y's from the n sums so that the other $n-k$ terms in the product are x's.
- Therefore, the coefficient of $x^{n-k}y^k$ is $\binom{n}{k}$.